Experiment-22

Aim

To perform image sharpening using a Laplacian mask with a positive center coefficient and enhance the edges and fine details of the image.

Procedure

1. Select the input image.
2. Read the image using OpenCV.
3. Convert the image into grayscale.
4. Define the Laplacian sharpening mask with a positive center coefficient.
5. Apply the mask to the image using convolution.
6. The positive center coefficient emphasizes the current pixel while the neighbouring pixels are subtracted.
7. This enhances edges and fine details in the image.
8. Display the original and sharpened images.
9. Compare the images and observe the sharpening effect.

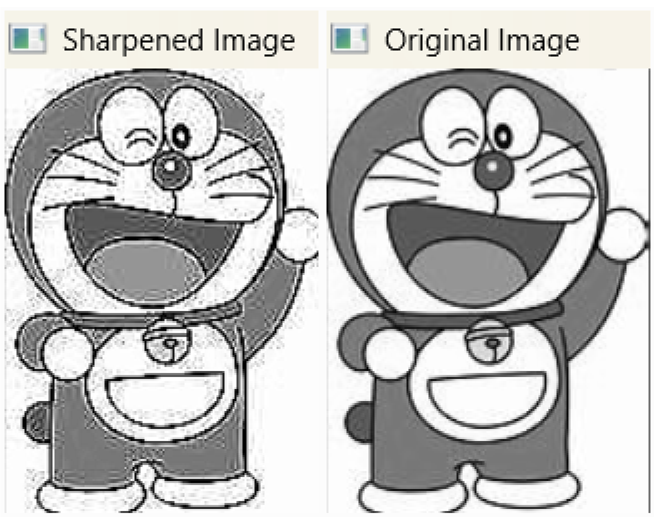
Python Code

```
import cv2
import numpy as np
img = cv2.imread("image.jpg")
if img is None:
    print("Error: Image not found")
    exit()
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
mask = np.array([
    [0, -1, 0],
    [-1, 5, -1],
    [0, -1, 0]
], dtype=np.float32)
sharpened = cv2.filter2D(gray, -1, mask)
```

```
cv2.imshow("Original Image", gray)
cv2.imshow("Sharpened Image", sharpened)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

Sample Input:

Output:



Results:

The image was successfully sharpened using the Laplacian mask with a positive center coefficient. The sharpened image shows enhanced edges, boundaries, and fine details compared with the original image.